

Phase 0 Benchmark Report — Nutrition RAG

Evaluated naive baselines against 100 lookup queries spanning common foodservice ingredients. Each query has expected keywords; a result is correct if any of its top-K matches contains ALL keywords (case-insensitive).

Headline metrics

Metric	Target	usda_api_direct	sql_fts5_local
Top-1 accuracy	≥ 0.90	75.0%	95.0%
Top-5 recall	—	76.0%	95.0%
Top-10 recall	≥ 0.95	77.0%	96.0%
MRR	—	0.757	0.951
Median match rank	1	1	1
Latency p50 (ms)	< 1000	4.2	861.9
Latency p95 (ms)	< 3000	10935.6	1094.1
Queries w/ no match	0	23	4

Scope decision

Recommendation: SCOPE DOWN. At least one baseline already hits the targets (top-1 95.0%, top-10 96.0%). The full RAG pipeline would buy little. Ship the better baseline as a thin LLM-wrapped service.

Failing queries (top 10 examples)

usda_api_direct — queries with NO match in top 10 (23)

- Q001 chicken breast raw (expected: ['chicken', 'breast', 'raw'])
- Q007 tuna canned (expected: ['tuna', 'canned'])
- Q008 pork chop raw (expected: ['pork'])
- Q009 shrimp raw (expected: ['shrimp', 'raw'])
- Q014 milk whole (expected: ['milk', 'whole'])
- Q016 butter unsalted (expected: ['butter'])
- Q023 strawberries raw (expected: ['strawberr'])

- Q026 avocado raw (expected: ['avocado', 'raw'])
- Q029 pineapple raw (expected: ['pineapple'])
- Q032 spinach raw (expected: ['spinach', 'raw'])

sql_fts5_local — queries with NO match in top 10 (4)

- Q010 bacon raw (expected: ['bacon'])
- Q044 oatmeal cooked (expected: ['oat'])
- Q074 black pepper ground (expected: ['pepper', 'black'])
- Q094 corn on the cob (expected: ['corn'])